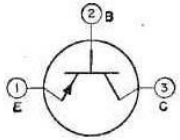
TRANSISTOR

2N2613



Germanium p-n-p type used in a wide variety of small-signal and low-power applications in high-quality af-amplifier equipment. It is a low-noise high-gain type intended primarily for use in the input and low-level stages of equipment having stringent performance requirements at low idling-current levels (0.3 to 0.7 milliampere). It can also be used in the input stages of phonograph amplifiers using either ceramic or magnetic pickups, tape recorders and players, microphone amplifiers, and other similar applications. It features a high small-signal forward current-transfer ratio, excellent linearity over the entire range of collector current, high cutoff frequency, low saturation currents, and uniform gain characteristics over the entire audio-frequency spectrum. JEDEC No. TO-1 package; outline 4, Outlines Section.

MAXIMUM RATINGS

Collector-to-Base Voltage	−30 max	volts
Collector-to-Emitter Voltage (with external base-to-emitter resistance = 10000 ohms)	−25 max	volts
Emitter-to-Base Voltage	−25 max	volts
Collector Current	−50 max	ma
Emitter Current	50 max	ma
Transistor Dissipation:		
At ambient temperatures up to 55°C	120 max	mw
At ambient temperatures above 55°C	Derate 2.6 mw/°C	
Temperature Range:		
Operating (junction) and Storage	−65 to 100	°C
Lead Temperature (for 10 seconds maximum)	255 max	°C

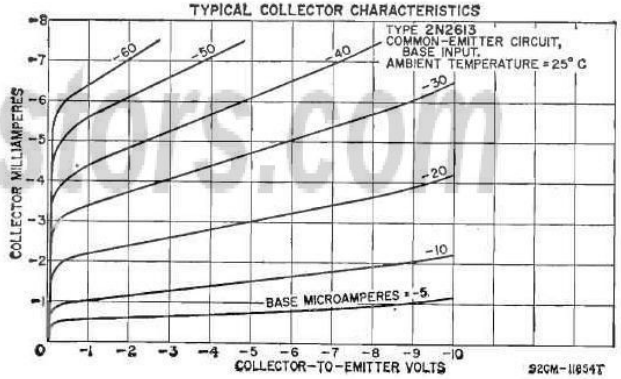
CHARACTERISTICS

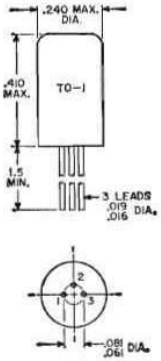
Collector-to-Base Breakdown Voltage (with emitter-to-base volts = −2 and collector ma = −0.05)	−30 min	volts
Collector-to-Emitter Breakdown Voltage (with collector ma = −1 and external base-to-emitter resistance = 10000 ohms)	−25 min	volts
Emitter-to-Base Breakdown Voltage (with emitter ma = −0.05 and collector current = 0)	−25 min	volts
Collector-Cutoff Current (with collector-to-base volts = −20 and emitter current = 0)	−5 max	μa
Emitter-Cutoff Current (with emitter-to-base volts = −20 and collector current = 0)	−7.5 max	μa
Extrinsic Base-Lead Resistance (measured at 20 Mc with collector-to-emitter volts = −4.5 and collector ma = −0.5)	300	ohms

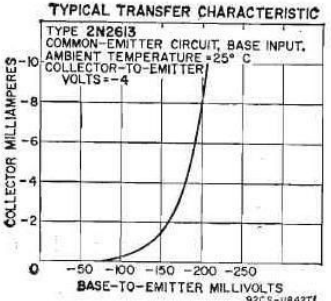
In Common-Emitter Circuit

Small-Signal Forward Current-Transfer Ratio (with collector-to-emitter volts = −4, colector ma = −0.5, and frequency = 1 kilocycle)	120 min	
Small-Signal Forward-Current Transfer-Ratio Cutoff Frequency with collector-to-emitter volts = −4.5 and collector ma = −0.5)	10 typ	Mc
Noise Figure (with collector-to-emitter volts = −4.5, collector ma = −0.5, generator resistance = 1000 ohms, circuit bandwidth = 1.1 kilocycle, and signal frequency = 1 kilocycle)	4 max	db
Collector-to-Base Feedback Capacitance (with collector-to-emitter volts = −4.5 and collector ma = −0.5)	10	pf
Equivalent RMS Noise Input Current for the 20- to 20000-cps frequency range (with collector-to-emitter volts = −4.5, collector ma = −0.5, and external base-to-emitter resistance = 50000 ohms)	0.001* max	μa

* At ambient temperatures above 25°C, value may be higher.







http://alltransistors.com